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Psychosocial aspects of *RUNX1*-familial platelet disorder in adolescents and young adults: "The fear of knowing it could happen sometime down the road"

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Abstract

Germline pathogenic variants in the *RUNX1* gene lead to the condition *RUNX1*-familial platelet disorder (*RUNX1*-FPD). This condition is associated with a 30%–50% lifetime risk of hematologic malignancies, including acute myeloid leukemia. In addition to physical manifestations including prolonged bleeding and easy bruising, individuals with *RUNX1*-FPD face profound psychosocial challenges. Individuals 18–39 years old may particularly struggle in their experiences with *RUNX*-FPD as they navigate developmental milestones. This descriptive, cross-sectional study includes participants aged 18–39, enrolled in the NIH RUNX1 Natural History Study (19-HG-0059). Each participant completed a structured psychosocial self-report assessment that included demographic factors, patient-reported outcomes on mental health symptomatology, psychiatric treatment history, and traumatic stress symptomatology. Open-ended questions asked participants to identify the most difficult part of living with *RUNX1*-FPD and how *RUNX1*-FPD has impacted their lives. Twenty-two participants completed the measures and were included in the analyses. Participants' mean age was 31 years (SD = 5.1). Fifty percent of participants reported higher than average anxiety scores, 59% reported they had been treated by a mental health professional, and 64% reported traumatic stress symptomatology. Living with the chronic uncertainty of cancer development and overall distress from recommended lifelong monitoring processes emerged as themes from the illustrative exemplar quotes. With this study, we describe the significant psychosocial and mental health challenges that emerging adults with *RUNX1*-FPD face. Access to psychological counseling, peer support groups, and genetic counseling can help individuals process their risk, manage anxiety, and make informed decisions. More research is needed to explore effective coping strategies, resilience-building techniques, and patient-centered resources to

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improve quality of life for those living with a hereditary predisposition to hematologic malignancies.

KEYWORDS
adolescents and young adults, cancer, hereditary cancer predisposition, lived experience, mental health, psychosocial

1 | INTRODUCTION

Germline pathogenic variants in the *RUNX1* gene lead to a condition called *RUNX1*-familial platelet disorder with associated myeloid malignancy (*RUNX1*-FPDMM, or *RUNX1*-FPD) (Cunningham et al., 2023). This condition is characterized by quantitative and qualitative platelet defects, along with an approximately 30%–50% lifetime risk of developing a hematologic malignancy (HHM) (Deuitch et al., 2021). *RUNX1*-FPD is one of a growing number of known hereditary predispositions to HHM (Stewart et al., 2025). In addition to physical manifestations such as abnormal bleeding and bruising, individuals with *RUNX1*-FPD may face profound psychosocial challenges. To date, there has been no formal evaluation of the psychosocial needs of individuals with *RUNX1*-FPD. Anecdotally, patients frequently express concerns surrounding the chronic uncertainty of cancer and bleeding risk along with the recommended lifelong monitoring for leukemia development through routine blood draws and bone marrow biopsies. Together, this can lead to pervasive anxiety, hypervigilance about symptoms, and overall heightened distress.

The psychosocial effects of living with a genetic predisposition to cancer can extend beyond the individual (Ringwald et al., 2016; Ross et al., 2017). Adolescents and young adults (AYA) ages 18–39 years may particularly struggle with such predispositions as they navigate their relationships, parenthood, and employment (Hoskins et al., 2021; Young et al., 2017). *RUNX1*-FPD is an autosomal dominant condition, and the 50% chance of passing the condition on to a child may additionally present complex emotional considerations in this particular age group (Cunningham et al., 2023).

Predictors of distress in individuals with *RUNX1*-FPD may vary depending on family history of malignancy and related deaths, presence of physical manifestations, and other psychosocial considerations (e.g., social support, psychiatric diagnoses). Given the wide range of psychosocial effects, particularly in the emerging adulthood age span, it is essential to investigate the unique lived experiences of individuals with *RUNX1*-FPD.

As there have been no studies examining the psychosocial experience, psychosocial functioning, or health- related quality of life of individuals living with *RUNX1*-FPD, we aimed to assess and describe these variables among a cohort of AYAs ages 18–39 living with a *RUNX1* germline variant, who may face age-specific vulnerabilities. There is growing recognition of the need for mental health care in hereditary cancer predispositions (Hanania et al., 2025). By assessing the lived experience among AYAs with *RUNX1*-FPD, recommendations can be made for the involvement of mental health care in the comprehensive medical care of these individuals.

What is known about this topic

To date, there has been no formal evaluation of the psychosocial needs of individuals with *RUNX1*-FPD, and there is growing recognition of the need for mental health care in hereditary cancer predispositions.

What this paper adds to the topic

This study provided an opportunity to better understand the psychosocial experiences of young adults living with *RUNX1*-FPD.

Plain Language Summary

RUNX1-FPD is a rare inherited condition caused by changes in the RUNX1 gene that can lead to bleeding problems and an increased risk of blood cancers like leukemia. Along with these physical health risks, individuals may experience significant anxiety and emotional distress related to lifelong cancer monitoring and uncertainty about their health.

2 | METHODS

2.1 | Research protocol

The natural history study *Longitudinal studies of patients and families with RUNX1/FPDMM caused by RUNX1 germline variants or FPDMM-like conditions* (NCT03854318) was launched at the National Human Genome Research Institute (NHGRI) in 2019 following NIH IRB approval to advance our understanding of the clinical, biological, and genomic features of *RUNX1*-FPD (Cunningham et al., 2023).

2.2 | Human studies and informed consent

Informed consent was obtained from all enrolled participants and all procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Declaration of Helsinki, as revised in 2000. Participants in the study undergo clinical and genomic evaluation through both in-person and remote participation. Study participants were invited to complete a survey assessing psychosocial measures via REDCap (Harris et al., 2009). Participants had the option of completing the measures remotely or during any in-person visit to the National Institutes of Health (NIH). Both adult and pediatric patients completed self-report measures and parents/caregivers of children or young adults of all ages completed proxy measures. For this paper, we are presenting the data of the emerging adult participants (ages 18–39).

2.3 | Measures

2.3.1 | Demographics

Participants' age, sex, race, ethnicity, and education levels were recorded.

2.3.2 | Psychosocial assessment interview

Due to the lack of a specific standardized instrument assessing how *RUNX1*-FPD impacts quality of life, a structured self-report assessment was designed for the study. A version of a structured psychosocial assessment was originally developed for other rare conditions at the NIH including Gastrointestinal Stromal Tumor and Medullary Thyroid Carcinoma (Lockridge et al., 2022; Wiener et al., 2012). We adapted the psychosocial assessment for the *RUNX1*-FPD cohort after conducting a literature review related to *RUNX1*-FPD and following discussions with experts caring for these patients. The assessment was then piloted and refined with three participants with *RUNX1*-FPD. The assessment contains items covering demographic factors, family stressors, general health, psychosocial concerns, psychiatric history, self-identified needs, expectations regarding disease outcome, and interest in a range of possible psychosocial services. Participants were asked open-ended questions to identify the most difficult part of living with *RUNX1*-FPD and how the condition has impacted their lives. The open-ended questions were also reviewed by medical and nursing staff experienced in the care of persons living with *RUNX1*-FPD. All psychosocial measures were built into the NIH REDCap system (Harris et al., 2009).

2.3.3 | Perceptions of physical health and pain

The study questionnaire inquired about other symptoms that have been problematic for the participant in the past year. Participant response options included fatigue, weakness, shortness of breath, bleeding, skin issues, bruising, frequent infections, abdominal pain, and memory loss. Response options included "Not at all," "Sometimes," "Often," and "All the time."

2.3.4 | Traumatic stress symptomatology

The psychosocial assessment asked participants how often they may or may not have experienced traumatic stress symptomatology, including nightmares, racing heart, and avoiding places when reminded of their *RUNX1*-FPD diagnosis. Response options included "Not at all," "Sometimes," "Often," and "All the time."

2.3.5 | Mental health treatment characteristics

Participants were asked a series of questions related to their history of mental health treatment, including if they had ever been treated by a mental health professional and if they had ever been prescribed psychotropic medications. Participants were also asked about family history of mental health symptoms, diagnoses, and treatment as well as *RUNX1*-FPD and cancer diagnoses.

2.3.6 | The patient-reported outcomes measurement information system (PROMIS)

The PROMIS is an NIH-funded initiative to develop and validate patient-reported outcomes (PROs) for clinical research and practice designed to evaluate and monitor physical, mental, and social health in adults and children (Bevans et al., 2014). Each PROMIS short form includes four to eight items, measures reported health outcomes in the past 7 days on a Likert-type scale, and takes approximately 2–3 minutes to complete. Included in the study was an assessment of Global Health (10-item) and several four-item measures assessing anxiety, depression, pain interference, fatigue, emotional support, family relationships, and belly pain. Participants completed the PROMIS anxiety, depression, pain interference, fatigue, and emotional support short forms. T-scores are categorized into a scale ascending from normal (average) to mild, moderate, and severe levels.

2.3.7 | Reasons for participation in the natural history study

Participants were asked to rate the extent to which various reasons influenced their decision to participate in the study. Reasons included wanting to contribute to scientific research, wanting to learn more about the disease, wanting to be connected with scientists on the leading edge of the disease, finances, and wanting to be followed closely for peace of mind. Response options were "Not at all," "A little bit," "A moderate amount," and "A lot."

2.3.8 | Benefit and burden

To assess the perceived benefit and burden of completing these measures within the natural history study, the Benefit and Burden Questionnaire was utilized (Pessin et al., 2008). This measure has been used successfully with other NIH studies (Wiener et al., 2015a, 2015b).

2.4 | Statistical analysis

In this cross-sectional study, descriptive analyses were conducted on quantitative data using R statistical software. Unless otherwise indicated, when items contained missing responses, the valid percent was reported. Illustrative exemplary quotes were chosen from the responses provided to the open-ended, free-text narrative questions. These responses were reviewed by two authors (OK, NTD) to identify common themes. Responses that were determined to fall under one or more thematic categories were categorized under all applicable themes.

3 | RESULTS

3.1 | Demographic characteristics

Twenty-eight participants in the total cohort were ages 18–39 years and eligible to receive the psychosocial assessment. Twenty-two of these participants completed all measures included in this study (79%). To control for differences based on support provided at NIH, we only utilized data from participants' first visit to the clinical center. The participants' mean age was 31 years (SD = 5.1) and was approximately evenly split between male (45%) and female (55%) participants. The majority were White (95%), non-Hispanic or Latine (95%), and highly educated with a college degree or graduate professional/graduate school education (86%). Additional demographic characteristics of participants are shown in Table 1. The total cohort of the natural history study is closely split between male (47%) and female (53%). The majority are White (94%) and non-Hispanic or Latine (87%). Thus, the present sample's demographic characteristics are relatively reflective of the total cohort.

TABLE 1 Participant demographics (*N* = 22).

Variable	Category	*n*	%
Sex	Male	10	45
	Female	12	55
Age	20–30	8	36
	31–38	14	64
Ethnicity	Hispanic/Latine	1	5
	Not Hispanic/Latine	21	95
Race	White	21	95
	Multiracial	1	5
Education	Less than high school	0	0
	Some college/vocational school	3	14
	Some professional/graduate school	0	0
	Graduated high school/GED	3	14
	Graduated college/vocational school	11	50
	Graduate of professional/graduate school	5	23

Note: Percentages are rounded to the nearest whole number. Percentages are based on the total sample (*N* = 22).

3.2 | Perceptions of physical health and symptomatology

One quarter of participants reported experiencing pain "a few days a week" or "every day." Over half of the participants rated their general health as "very good" or "excellent." Twenty-seven percent endorsed fatigue as being problematic "a lot," and 18% reported skin issues as also being problematic "a lot" (Table 2).

3.3 | Mental health treatment characteristics

Over half of the participants reported being treated by a mental health professional. The mean age at time of first treatment is 21.2 years (SD = 7) and 62% of the participants were still under the care of a mental health professional at the time of evaluation. Of those who had been under the care of a mental health professional, a majority reported the reason for treatment is to manage anxiety symptoms. Other reported reasons for treatment include depression and grief of loved ones who died of cancer.

Half of participants reported being prescribed medication for anxiety. The average age at first treatment was 25 years (SD = 5). Other than anxiety and generalized anxiety disorder, reported reasons for medication included management for panic disorder, distracting thoughts, and postpartum symptomatology. Cancer-related mental health symptoms were also mentioned, as 18% of the sample reported a previous cancer diagnosis. A majority were still taking anxiety medication at the time of evaluation (Table 3).

Over one-quarter of participants reported being prescribed medication for depression. The average age at the time of first treatment was 20 years (SD = 5). Three participants reported they were still taking this medication at the time of evaluation (Table 3). Less common was being prescribed medication for attention problems.

3.4 | Traumatic stress symptomatology

When reminded of their *RUNX1*-FPD diagnosis, over half of participants reported some traumatic stress symptomatology. Almost half of participants endorsed sweating, trembling, or racing heartbeat, and almost one third endorsed bad dreams or nightmares. Nineteen percent of participants endorsed avoiding going to the doctor or other places that remind them of their diagnosis (Table 4).

3.5 | PROMIS measures

All participants completed all PROMIS measures (Table 5). Half of the sample reported higher than average anxiety scores, with 23% falling into the mild category, 18% in the moderate category, and 9% in the severe category.

TABLE 2 Symptomatology ($N = 22$).

Question	Not at all, *n* (%)	A little, *n* (%)	A moderate amount, *n* (%)	A lot, *n* (%)
Fatigue	5 (23)	5 (23)	6 (27)	6 (27)
Weakness	11 (50)	6 (27)	3 (14)	2 (9)
Shortness of breath	11 (50)	10 (45)	1 (5)	0 (0)
Bleeding	9 (41)	7 (32)	4 (18)	2 (9)
Skin issues	5 (23)	8 (36)	5 (23)	4 (18)
Bruising	4 (18)	9 (41)	6 (27)	3 (14)
Frequent infections	19 (86)	3 (14)	0 (0)	0 (0)
Belly Pain	14 (64)	5 (23)	3 (14)	0 (0)
Memory	10 (48)	6 (29)	4 (19)	1 (5)

Note: Percentages are rounded to the nearest whole number. Percentages are based on the total sample ($N = 22$).

TABLE 3 Psychiatric medications ($N = 22$).

Question	Yes, *n* (%)	No, *n* (%)	Age at first treatment, years ($M \pm SD$)	Total (*n*)
Have you ever been prescribed medication for anxiety?	11 (50)	11 (50)	25 ± 5	22
Are you still taking medication for anxiety?	8 (73)	3 (27)	n/a	11
Have you ever been prescribed medication for depression?	6 (27)	16 (73)	20 ± 5	22
Are you still taking medication for depression?	3 (50)	3 (50)	n/a	6

Note: Percentages are rounded to the nearest whole number. Percentages are based on the total sample ($N = 22$).

TABLE 4 Mental health characteristics: traumatic stress symptomatology ($N = 22$).

Question	Not at all, *n* (%)	Sometimes, *n* (%)	Often, *n* (%)	All the time, *n* (%)
Bad dreams/nightmares	15 (68)	7 (32)	0 (0)	0 (0)
Sweating, trembling, racing heartbeat	13 (59)	7 (32)	2 (9)	0 (0)
Avoided going to doctor/other places	18 (82)	3 (14)	0 (0)	1 (5)

Note: Percentages are rounded to the nearest whole number. Percentages are based on the total sample ($N = 22$).

Twenty-three percent of the sample reported higher than average depression scores, with 5% in the mild category and 18% in the moderate category. No participants reported levels of severe depression. Thirty-seven percent of the sample reported higher than average pain interference levels, with 23% reporting mild scores and 14% reporting moderate scores. Twenty-eight percent of the sample reported higher than average levels of fatigue, with 14% reporting mild scores, 9% reporting moderate scores, and 5% reporting severe levels (Table 5).

3.6 | Open-ended questions: "The Fear of Knowing it Could Happen Sometime Down the Road"

Four consistent themes emerged from the open-ended question, "What is the most difficult part of living with *RUNX1*?" including living with uncertainty and fear, family-related concerns, feeling alone, and living with ongoing physical symptoms. Over half of participants endorsed living with uncertainty and fear. More than one-quarter of participants reported family-related concerns, including anticipatory grief with family members potentially dying of cancer and uncertainty about how/if family members are affected. Participants were also asked, "How has *RUNX1* affected your life?" Four common themes from these answers include anxiety/stress/sadness, isolation, being more vigilant with their health, and family-related concerns. Half of participants reported anxiety/stress/sadness. Exemplar quotes can be found in Table 6.

Participants who endorsed experiencing pain were asked a follow-up open-ended question of how this pain interferes with the activities in their lives. Of these five participants who reported pain, nearly all endorsed being limited or slowed down by their pain. Forty-one

TABLE 5 PROMIS measures (*N* = 22).

Symptom	Average, *n* (%)	Mild, *n* (%)	Moderate, *n* (%)	Severe (%)
Anxiety	11 (50)	5 (23)	4 (18)	2 (9)
Depression	17 (77)	1 (5)	4 (18)	0 (0)
Pain Interference	14 (64)	5 (23)	3 (14)	0 (0)
Fatigue	16 (73)	(14)	2 (9)	1 (5)

Note: Percentages are rounded to the nearest whole number. Percentages are based on the total sample (*N* = 22).

TABLE 6 Open-ended questions.

Question	Exemplar quotes
What is the most difficult part of living with *RUNX1*-FPD?	"The fear of knowing it could happen sometime down the road in my life but not that it's a certainty." "Having to worry constantly and not knowing what could happen in the future." "Thinking that anytime something goes wrong medically/health wise it might have something to do with RUNX1 but not really knowing." "Menta(l). Wondering if I will die from RUNX1 as several family members already have." "Trying to understand how it relates to all my other health issues and what may happen in the future." "…the hardest part has probably been just trying to understand what it means for me and any family members."
How has *RUNX1*-FPD affected your life?	"Every time I bruise I am reminded of what might happen to me. I am also now unsure if I should have children and if I have enough money for IVF. I also wonder if and if I am going to be able to live a long happy life. During happy activities I now think if I get cancer and die at least I can be happy in this moment. When I see someone in the runx1 Facebook group passing away or going through a hard time it ruins my day and makes me very sad. It also makes me sad to know my dad, and 3 cousins also have this. My aunt and grandma who had runx1 both died from blood cancer, so it constantly weighs on my mind. I feel alone and sad, and also feel sad and scared for my parents and my siblings." "Heightened sense of my own mortality." "As a child having my school think I have problems at home because my mother and I had so many bruises was confusing. I lost a cousin to AML when she was only 22. Few years after that I lost my mother to AML. Wondering if that will be the same fate for me. And wondering how I can make my daughter's life as normal as possible. And wondering if having more children is something I want to do!" "Increased stress and urgency to reach goals/dreams." "Worry, anxiousness."

percent of participants reported their pain interfered with their mood and were asked a follow-up open-ended question to further describe the impact of pain. Three common themes emerged, including irritation/frustration, feeling drained/fatigued, and general negative impact on mood. Over half described irritation/frustration, and one-third described feeling drained/fatigued.

Finally, participants were asked to describe how the information they have been provided throughout their care at NIH has or has not changed how they think about their future health care. Fifteen (68%) participants responded. All of these individuals described being more vigilant about their health and staying informed about information on *RUNX1*-FPD. Some described concerns moving forward about health insurance and medical financial responsibilities.

3.7 | Interest in supportive resources

At the end of the psychosocial assessment, participants were asked if they would be interested in learning more about the supportive resources (nutritional guidance, fertility counseling, support group, etc.) available to them. All participants endorsed being interested in at least one of the resources.

3.8 | Reasons for participation in the natural history study

Participants were asked to rate the extent to which various reasons influenced their decision to participate in the study. Response options were "Not at all," "A little bit," "A moderate amount," and "A lot." In response to statements to the following prompt, "What were the reasons you chose to participate in this study?" Seventy-one percent of participants reported helping others as the strongest reason for participating in the study. Sixty-seven percent reported that they wanted to learn more about *RUNX1* and stay connected to science.

3.9 | Benefit and burden

When asked if completing the set of psychosocial questionnaires was difficult, 73% responded "Not at all," 13% responded "A little bit," 9% responded "Somewhat," and 5% responded "Quite a bit." For those (28% of total sample) who reported the questions were difficult, they were asked a follow-up question of why they found it difficult. Responses include, "The questions were too personal or upsetting to me," "I wasn't feeling well when completing them," "It was too long/had too many questions."

When asked if completing the set of psychosocial questionnaires was beneficial, 9% responded, "Not at all," 41% responded "A little bit," 27% responded "Somewhat," 18% responded "Quite a bit," and 5% responded "Very much." For those (91% of total sample) who said the questionnaires were helpful, they were asked a follow-up question of why they found it to be helpful. Responses include, "It made me feel good to help others/contribute to society," "It addressed issues that are important to me," "It was helpful or a relief to talk about my illness," "It helped pass the time/kept my mind busy."

4 | DISCUSSION

This study provided an opportunity to better understand the psychosocial experiences of young adults living with *RUNX1*-FPD. This data suggests that those with *RUNX1*-FPD may face additional psychosocial challenges associated with the developmental circumstances typically experienced in young adulthood. In this sample, 62% of individuals reported receiving mental health treatment, which is notably higher than national treatment rates (National Center for Health Statistics, 2024). While we are unable to draw a direct association between their condition and their experiences of anxiety, it is important to consider that the burdens associated with living with *RUNX1*-FPD may add to the stress and anxiety already experienced by young adults during this transitional time in their lives. Family-related concerns and employment and insurance needs that support the lifelong monitoring process and potential need for medical treatment were commonly identified stressors among this cohort. Thus, psychological care tailored to these age-specific factors may be beneficial for individuals with *RUNX1*-FPD and other genetic variants requiring similar medical treatment and follow-up.

Participants in this study endorsed higher rates of pain and fatigue than the general population (Rikard et al., 2023; MMWR, CDC 2023). Skin issues related to bruising or eczema were also reported, which are found more frequently in individuals with *RUNX1*-FPD (Deuitch et al., 2021). However, pain and fatigue have not explicitly been linked to the condition in the past. Fatigue may be a consequence of living with a chronic condition and the mental toll that can take. Future research is needed to explore the nature of the pain reported (along with fatigue and skin issues) and how this may be related to *RUNX1*-FPD.

Interestingly, most participants rated their general health as either very good or excellent yet reported mental health symptomatology and health-related anxiety. Concern about the uncertainty of future disease may have a negative impact on individuals living with *RUNX1*-FPD, and many participants expressed feelings suggestive of anticipatory grief—especially for those with an extensive family history of leukemia. This calls attention to the need for integrated psychosocial support that is tailored to both disease and age-specific factors in the experience of living with *RUNX1*-FPD.

Participants reported having avoided medical appointments or other places that remind them of *RUNX1*-FPD, experiencing nightmares and somatic responses (e.g. sweating, racing heartbeat) when reminded of their condition, suggesting that psychological symptoms may adversely impact medical care and worsen outcomes. Conversely, individuals in this study described that having a concrete monitoring plan made them feel more in control. It is essential that the need for psychosocial care is recognized in the comprehensive medical care for individuals living with a hereditary predisposition to malignancies in order to help support medical decision making, foster adaptive coping in the face of chronic anxiety and uncertainty, manage the use of psychiatric medication, enhance family communication, and guide future family planning (Hanania et al., 2025; Stewart et al., 2025).

4.1 | Practice implications and research recommendations

Genetic counselors and mental healthcare providers including psychologists, psychiatrists, and social workers can help to support the unique psychosocial needs of individuals with *RUNX1*-FPD. The roles of both mental health professionals and genetic counselors in the multidisciplinary medical team can help individuals process their risk, manage anxiety, and make informed decisions. Further, the involvement of these providers should be ongoing through the lifelong monitoring process, as psychosocial needs may vary by age group.

Genetic counselors are particularly well positioned to provide psychosocial support within the context of hereditary cancer and hematologic predisposition syndromes because of their specialized training in both medical genetics and counseling. While the extent of long-term follow-up psychosocial support provided may vary by practice setting, genetic counselors can help connect patients with supportive resources and facilitate referrals when additional psychosocial care is needed. Specifically, genetic counselors can help individuals regain a sense of control amidst the uncertainty associated with disease development by facilitating timely communication of bone marrow biopsy or genetic testing results and providing proactive, anticipatory guidance regarding potential outcomes, particularly those associated with leukemia risk. For this age group in particular, discussions surrounding family planning and reproductive options, including in vitro fertilization with preimplantation genetic testing (PGT-M), may also help individuals and families feel more empowered and informed in managing their condition (Stewart et al., 2025).

Considering the prevalence of mental health diagnoses and reported anxiety among this population, psychologists and social workers can offer psychotherapeutic interventions, coping strategies, and psychosocial support amid distressing disease-related uncertainty. Further, psychiatrists can play a critical role in the psychological care of these individuals through psychotropic medication recommendations and management.

This data also supports the need for more formal consensus guidelines for the clinical management of *RUNX1*-FPD involving a multidisciplinary team, as participants cited uncertainty about management as being a stressor. Additionally, more research is needed to explore effective coping strategies, resilience building techniques, and patient-centered resources to improve quality of life for those living with hereditary hematologic malignancies. Further research on the psychosocial impacts of living with *RUNX1*-FPD and effective coping strategies is needed to inform the development of tailored psychotherapeutic interventions.

5 | LIMITATIONS

This study is not without limitations. Because this is a descriptive study, definitive conclusions and associations cannot be drawn regarding the diagnosis of *RUNX1*-FPD and psychosocial sequelae. Considering this transitional time in participants' lives, their experiences of anxiety and depression may be solely or additionally attributable to other events separate from their diagnosis. Generalization of the findings for all persons living with *RUNX1*-FPD and inference may also be limited, as the sample size is relatively small and predominantly composed of white, non-Hispanic participants. Other demographic groups could experience distinct psychosocial challenges that are not fully captured in this cohort. Additionally, there is likely some degree of ascertainment bias. Nearly all probands were referred to in the study following a diagnosis of *RUNX1*-FPD, which may result in an overrepresentation of individuals with more severe clinical manifestations, greater access to healthcare, and those who are more medically proactive or interested in research participation. Ongoing efforts aim to expand recruitment through large population-based datasets and to enhance study accessibility with the goal of achieving a more representative understanding of the psychosocial needs of individuals affected by *RUNX1*-FPD. Lastly, we did not collect data on the time since the participants' diagnosis. Future longitudinal research should capture how the recency of diagnoses may influence the stress and uncertainty that individuals with *RUNX1*-FPD face.

6 | CONCLUSION

Young adults with *RUNX1*-FPD may face significant psychosocial impacts that affect their mood, daily life activities, and familial concerns. Given the experiences of anticipatory grief, anxiety, and fear of a cancer diagnosis or relapse among this cohort, coupled with the typical anxiety and stressors associated with this age group, attention to the psychosocial needs of individuals with *RUNX1*-FPD is important. Genetic counselors can implement the recommended strategies to reduce patient uncertainty and ensure they are adequately informed about reproductive planning options. Further, the involvement of psychologists, psychiatrists, and social workers can be beneficial for this patient population to manage psychological distress secondary to their *RUNX1*-FPD diagnosis.

AUTHOR CONTRIBUTIONS

Olivia Kearns: Data curation; investigation; methodology; writing – original draft; writing – review and editing. **Natalie T. Deuitch:** Conceptualization; data curation; investigation; methodology; project administration; writing – original draft; writing – review and editing. **Kathleen Craft:** Conceptualization; data curation; investigation; methodology; project administration; writing – review and editing. **David J. Young:** Resources; supervision; writing – review and editing. **Paul P. Liu:** Resources; supervision; writing – review and editing. **Lori Wiener:** Conceptualization; data curation; investigation; methodology; project administration; resources; supervision; writing – original draft; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

Authors Olivia Kearns, Natalie T. Deuitch, Kathleen Craft, David J. Young, Paul P. Liu, and Lori Wiener declare that they have no conflict of interest. The contributions of the NIH author(s) were made as part of their official duties as NIH federal employees, are in compliance with agency policy requirements, and are considered Works of the United States Government. However, the findings and conclusions presented in this paper are those of the author(s) and do not necessarily reflect the views of the NIH or the US Department of Health and Human Services. Study data were collected and managed using REDCap electronic data capture tools hosted within the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). Consulting support was provided by the Biomedical Translational Research Information System (BTRIS). Both NIDDK and BTRIS are part of the National Institutes of Health (NIH). REDCap (Research Electronic Data Capture) is a secure, web-based application designed to support data capture for research studies. REDCap was developed and is licensed by Vanderbilt University.

DATA AVAILABILITY STATEMENT

Data are available on request due to privacy/ethical restrictions.

ETHICS STATEMENT

Human studies and informed consent: Informed consent was obtained from all enrolled participants and all procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Declaration of Helsinki, as revised in 2000. Participants on the study undergo clinical and genomic evaluation through both in-person and remote participation.

Animal studies: Animal studies were not applicable.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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